

Faculty Vitae: Meenesh Singh

1. Education

B.S.: B.E., Chemical Eng., Sardar Patel University, Gujarat, India, August 2001-05, GPA: 8.41/10

M.S.: M.Tech., Chemical Eng., Indian Institute of Technology Bombay, India, August 2006-08
GPA: 9.18/10

Ph.D.: Chemical Engineering, Purdue University, USA, Jan 2009-March 2013 GPA: 4.0/4.0

2. Academic Experience

Full Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2025 – Present
Joint Appointment Argonne National Laboratory	April 2024 – present
Visiting Faculty Department of Chemical Engineering, Indian Institute of Technology Roorkee	Jan 2023 – present
Associate Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2023 – July 2025
Associate Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2016 – July 2023

3. Professional Experience

Chief Technology Officer eN-RAMPS LLC.	July 2021 – present
Affiliate Scientist Joint Center for Artificial Photosynthesis, Lawrence Berkeley National Lab	Aug 2016 – Nov 2017
Postdoctoral Scholar Joint Center for Artificial Photosynthesis, U C Berkeley, Lawrence Berkeley National Lab Title: Artificial Photosynthesis for CO ₂ Reduction Supervisor: Alexis T. Bell, Adam Z. Weber	Jan 2014 – Aug 2016
Postdoctoral Scholar Joint Center for Artificial Photosynthesis, U C Berkeley, Lawrence Berkeley National Lab Title: Analysis of Solar-Fuel Generators Supervisor: Rachel A. Segalman	March 2013 – Jan 2014
Research Intern, Abbott Labs: Global Pharma R&D Title: Experimental Investigation of Crystal Morphology during Dissolution and Growth Supervisors: Nandkishor Nere, Hsien-Hsin Tung and Shailendra Bordawekar	May 2011 – July 2011
Graduate Research Assistant, Purdue University Title: Towards Control of Crystal Shape and Morphology Distributions in	Jan 2009 – March 2013

Crystallizers

Advisor: Doraiswami Ramkrishna

Research Assistant, IIT-B

July 2008 – Nov 2008

Title: Monte Carlo Simulations for Growth of Nanoparticles in Microemulsion

Supervisors: Anurag Mehra, Rochish Thaokar and Jayesh Bellare

Graduate Research Assistant, IIT-B

Jan 2007 – May 2008

Title: Cryogenic Grinding for Synthesis of Nanoparticles and Extraction of Protein and DNA from Cells

Advisors: Jayesh Bellare and Sandip Roy

4. Certifications or professional registrations

Editor	Chemical Engineering Research & Design (Elsevier)	2020 – Present
Associate Editor	Energy Lab (Lab Academic Press)	2022 - Present
Advisory Board	Challenges (MDPI)	2020 - Present
Guest Editor	Special issue on “Modeling and Control of Crystallization” in Processes	2019
Guest Editor	Special issue on “Electroreduction of Activated Nitrogen Compounds” in Communications Chemistry (Nature Portfolio Journal)	2024

5. Current membership in professional organizations

Member	American Institute of Chemical Engineers	2009 – Present
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6. Honors and awards

1.	Faculty Entrepreneur Leadership Program (FELP) Cohort Member	Aug 2025
2.	UIC’s Outstanding Global Scholar	Sep 2024
4.	UIC Proof-of-Concept Phase-II Award	Jan 2023
5.	UIC CTRI Award	May 2023
6.	UIC CoE Advising Award	Sep 2022
8.	UIC Proof-of-Concept Phase-I Award	Jan 2021
9.	UIC CTRI Award	May 2021
10	UIC CoE Research Award	May 2021
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11	George Klinzing Best PhD Award, AIChE	Nov 2015
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13	Faculty Lectureship Award, School of Chemical Engineering, Purdue University	March 2014
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14	AIChE Process Development Division Student Paper Award	Nov 2013
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15	Outstanding Graduate Student Research Award, College of Engineering, Purdue	March 2013

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- 16 AIChE Separation Division Graduate Student Research Award Oct 2012
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- 17 McDonnell Douglas Fellowship Fall 2012
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7. Service activities

- Member of 4 college-level committees
- AIChE Chem-E-Car and ChemE Cube advisor at UIC
- Editorial board member for 5 scientific journals
- Organized 34 technical sessions at international conferences

8. Selected Publications

1. Emani, Pavan S., Hisham Maddah, Arjun Rangoonwala, Songwei Che, Aditya Prajapati, Meenesh R. Singh, Dieter M. Gruen, Vikas Berry, and Sanjay K. Behura. "Organophilicity of Graphene Oxide for Enhanced Wettability of ZnO Nanorods," ACS Appl. Mater. Interfaces, 12 (35), 39772 – 39780, 2020.
2. Nishithan C. Kani, Aditya Prajapati, Brianna A. Collins, Jason D. Goodpaster, and Meenesh R. Singh, "Competing Effects of pH, Cation-Identity, H₂O Saturation, and N₂ Concentration on the Activity and Selectivity of Electrochemical Reduction of N₂ to NH₃ on Electrodeposited Cu at Ambient Conditions," ACS Catalysis, 10 (24), 14592 – 14603, 2020.
3. Ayankola O. Ayansiji, Anish V. Dighe, Andreas A. Linninger, and Meenesh R. Singh, "Constitutive Relationship and Governing Physical Properties for Magnetophoresis," PNAS, 117 (48), 30208-30214, 2020.
4. Aditya Prajapati, Brianna A. Collins, Jason D. Goodpaster, and Meenesh R. Singh, "Fundamental Insights into Electrochemical Activation of Methane towards Methanol over Transition Metal Oxides," PNAS, 118 (8), e2023233118, 2021.
5. Paria Coliaie, Manish S. Kelkar, Marianne Langston, Chengxiang Liu, Neda Nazemifard, Daniel Patience, Dimitri Skliar, Nandkishor K. Nere, and Meenesh R. Singh, "Advanced Continuous-Flow Microfluidic Device for Parallel Screening of Crystal Polymorphs, Morphology and Kinetics at Controlled Supersaturation," Lab on a Chip, 21 (12), 2333-2342, 2021.
6. John Cavin, Alireza Ahmadiparidari, Leily Majidi, Arashdeep Singh Thind, Saurabh N. Misal, Aditya Prajapati, Zahra Hemmat, Sina Rastegar, Andrew Beukelman, Meenesh R Singh, Kinga A. Unocic, Amin Salehi-Khojin, Rohan Mishra, "Two-Dimensional High-Entropy Transition Metal Dichalcogenides for Carbon Dioxide Electrocatalysis," Adv. Mater., 33, 2100347, 2021.
7. Nishithan C. Kani, Joseph Gauthier, Aditya Prajapati, Isha Bordawekar, Windom Shields, Mitchell Shields, Aayush R. Singh, Meenesh R. Singh, "Solar-Driven Electrochemical Synthesis of Ammonia with >10% Solar-to-Fuel Efficiency at Ambient Conditions," Energy Environ. Sci., 14, 6349-6359, 2021.
8. Aditya Prajapati, Rohan Sartape, Tomas Rojas, Naveen K. Dandu, Pratik Dhakal, Amey S. Thorat, Jiahua Xie, Ivan Bessa, Miguel T. Galante, Marcio H. S. Andrade, Robert T. Somich, Marcio V. Rebouças, Gus T. Hutras, Nathalia Diniz, Anh T. Ngo, Jindal Shah and Meenesh R.

- Singh, "Migration-assisted, moisture gradient process for ultrafast, continuous CO₂ capture from dilute sources at ambient conditions," *Energy Environ. Sci.*, 15, 680-692, 2022.
9. Anish V. Dighe, Prem K. R. Podupu, Paria Coliaie, and Meenesh R. Singh, "Three-Step Mechanism of Antisolvent Crystallization," *Cryst. Growth Des.*, 22 (5), 3119-3127, 2022.
 10. Aditya Prajapati, Rohan Sartape, Nishithan Kani, Joseph A. Gauthier, and Meenesh R. Singh, "Chloride-Promoted, High-Rate Ambient Electrooxidation of Methane to Methanol on a Patterned Cu-Ti Bimetallic Oxides," *ACS Catalysis*, 12 (22), 14321- 14329, 2022.
 11. Aditya Prajapati, Rohan Sartape, Miguel T. Galante, Jiahua Xie, Samuel Leung, Ivan Bessa, Marcio H. S. Andrade, Robert T. Somich, Marcio V. Rebouças, Gus T. Hutras, Nathalia Diniz, and Meenesh R. Singh, "Fully-Integrated Electrochemical System that Captures CO₂ from Flue Gas to Produce Value-Added Chemicals at Ambient Conditions," *Energy Environ. Sci.*, 15, 5105 - 5117, 2022.
 12. Anish V. Dighe, Prem K. R. Podupu, and Meenesh R. Singh, "Emulsification of Supersaturated Solutions Amplifies Induction Time Variation in Crystallization," *Cryst. Growth Des.*, 23 (9), 6290 – 6297, 2023
 13. Rohit Chauhan, Rohan Sartape, Rashmi Mishra, Jindal K. Shah, and Meenesh R. Singh, High-throughput Measurements of CO₂ Permeance and Solubility in Ionic Liquid Reveal a Synergistic Role of Ionic Interactions and Void Fractions, *Chem. Eng. J.*, 496, 153697, 2024.
 14. Ishita Goyal, Nishithan C. Kani, Samuel A. Olusegun, Sreenivasulu Chinnabattigalla, Rajan R. Bhawnani, Ksenija D. Glusac, Aayush R. Singh, Joseph A. Gauthier, and Meenesh R. Singh, "Metal Nitride as a Mediator for the Electrochemical Synthesis of NH₃," *ACS Energy Lett.*, 9, 4188-4195, 2024.
 15. Samuel A. Olusegun, Yancun Qi, Nishithan C. Kani, Meenesh R. Singh, and Joseph A. Gauthier, "Understanding activity trends in electrochemical dinitrogen oxidation over transition metal oxides," *ACS Catalysis*, 14, 16885-16896, 2024.
 16. Rajan R. Bhawnani, Rohan Sartape, Vamsi V Gande, Michael L. Barsoum, Elias M. Kallon, Roberto dos Reis, Vinayak P. Dravid, and Meenesh R. Singh, "Non-aqueous Electrochemical CO₂ Reduction to Multivariate C2-products over Single Atom Catalyst at Current Density up to 100 mA/cm²," *Small*, 21 (18), 2408010, 2024.
 17. Prem K.R. Podupu, Vamsi V. Gande, Ragavendra Hari, Akshay Korde, Manish S. Kelkar, Nandkishor K. Nere, Meenesh R. Singh, "Analytical solution to the discretized population balance equation for pure breakage with application to kernel identification," *Chem. Eng. Res. Des.*, 217, 295-303, 2025.
 18. Rohan Sartape, Rohit Chauhan, Venkata S.R.P. Yadavalli, Ishita Goyal, Ishaku Amos, Yancun Qi, Vamsi V. Gande, Abdul M. Sarkar, Ksenija D. Glusac, and Meenesh R. Singh, "Tetraethylammonium Cation Activates Fe for Selective Electroreduction of CO₂ to Oxalate," *ChemElectroChem*, e202500093, 2025.
 19. Rohit Chauhan, Huda Usman, Nitin Minocha, Mehdi Molaei, Tagbo H.R. Niepa, and Meenesh R. Singh, "Predictive Modeling and Experimental Validation of Magnetophoretic Delivery of Magnetic Nanocultures," *ACS Mater. Lett.*, 7, 2679 – 2685, 2025.
 20. Prem K.R. Podupu, Prince Verma, Ankit Dhakal, Rajan R. Bhawnani, Meagan Phister, Anish V. Dighe, Kevin H. Stone, Gaurav Giri, and Meenesh R. Singh, "Mechanistic Insights into Metal-Organic Framework Thin Film Growth from Microkinetic Analysis of In-Situ X-Ray Scattering Data," *Matter*, 102430, 2025.

21. Ishita Goyal, Najmin Isa, Vamsi V. Gande, Rohit Chauhan, and Meenesh R. Singh, “Stabilizing Calcium Nitride for Efficient, Long-Term Electrochemical Ammonia Synthesis,” PNAS, 2025.

9. Professional Development

- Awarded total project funding ~ \$9.2 MM and about \$3.7 MM to his lab.
- Research lab expenditures >\$500 K per year.
- Strong collaboration with 7+ multinational companies with continued funding support.
- Chief Technology Officer at eN-RAMPS LLC. (UIC Start-up)
- Actively involved in prototyping and scale-up of 4 technologies developed in PI’s lab.
- Raised ~\$1M in industrial funding to scale up technologies.